

Youshi (Tiger) Chen

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PROFESSIONAL SUMMARY

Mechanical engineering student at Stanford with hands-on experience in mechanical design, fabrication, and systems integration across autonomous UAVs, custom PCB housings, and full-stack competition robots. Combines rigorous CAD and prototyping fundamentals with end-to-end build experience from concept through validated prototype.

EDUCATION

Stanford University

Bachelor of Engineering, Mechanical Engineering.

Expected May 2030

Troy High School, Troy Tech Magnet Program

High School Diploma. GPA: 4.84.

2022-2026

Relevant Coursework: Multivariable Calculus, AP Physics C, Mechatronics Honors, Cambridge A-Level Design and Technology, AP Computer Science A

EXPERIENCE

Startup Co-Founder / Autonomous Drone Engineering Lead

2025 – 2026

Aircart

Brea, CA

- Engineered a heavy-lift UAV with a 5 kg payload capacity, validated through 20+ test flights with iterative structural improvements.
- Directed full-stack hardware development across structural integration, PID tuning, and flight validation, driving a 4-person team from concept to flight in 2 months.

Board Member, Electrical and Mechanical Team

2024 – 2026

FRC Robotics

Fullerton, CA

- Designed and assembled mechanical subsystems for a 100+ lb competition robot, integrating drivetrains, actuators, and structural components with a 20-person team.

Engineering Intern

Jun 2025 – Jul 2025

Atlas Assembly / Glenair

Arcadia, CA

- Designed assembly tooling in SOLIDWORKS and 3D-printed prototypes to improve assembly efficiency and technician access.
- Fabricated wire connector and fiber optic assemblies via precision crimping and termination to aerospace standards.

PROJECTS

3D Filament Recycler

2025 – 2025

Mechanical Design Project

- Designed and built a filament recycler converting failed PLA prints into reusable 1.75 mm spools, achieving diameter consistency within +/-0.05 mm across 5 design iterations.
- Iterated extrusion nozzle geometry, spooling arm, and feed mechanism to achieve target dimensional tolerances.

Badminton Racket Accelerometer PCB

2025 – 2025

PCB / Embedded Systems Project

- Designed a custom ESP32 PCB with an onboard IMU capturing 3-axis swing acceleration and impact force at 1000 Hz.

ACTIVITIES & LEADERSHIP

Competitive Badminton

2014-2026

Yonex-Sponsored Athlete

- Represented Team USA internationally; ranked #73 worldwide. Placed 1st at the 2025 Pan Am Junior Championships.

SKILLS

CAD / Design: SOLIDWORKS, Fusion 360, Onshape, parametric modeling, mechanical design, tolerance analysis

Fabrication: 3D printing, prototype testing, precision assembly, crimping, fiber optic cable assembly

Electrical / PCB: KiCad, schematic design, PCB layout, circuit design, power distribution, ESP32, IMU, sensors, motor controllers, soldering

Embedded Systems: firmware integration, PID tuning, sensor fusion, data acquisition

CERTIFICATIONS

- Certified SOLIDWORKS Design Associate, Dassault Systemes - Issued May 2026

AWARDS & HONORS

- 3-time Gold Presidential Volunteer Service Award
- National First-Generation Recognition Program